



Inheritance (OOP's) in Java:

1. Java, inheritance is an important pillar of Oop(Object Oriented Programing).
2. Parent and child relationship is called as inheritance.
3. Parent class is known as super class and child class is known as sub-class.
4. Object class is the topmost class in java.
5. The purpose of achieving inheritance is code reusability.
6. We can achieve inheritance through the “**extends**” keyword.
7. Java class does not support multiple inheritance so hybrid inheritance is not possible.
8. It is the mechanism in java by which one class is allowed to inherit the features (fields and methods) of another class.



9. In java, inheritance means creating new classes based on existing ones.
10. Java inheritance is also known as “**Is-A Relationship**”.

Syntax of Inheritance:

The extends keyword is used for inheritance in java. Using the extends keyword indicates you are derived from an existing class. In other words, “**extends**” refers to increased functionality.

```
public class Derived_Class extends Base_Class
{
    // methods and Fields
}
```

Where - Derived_Class → Child Class (Sub-Class)
Base_Class → Parent Class (Super-Class)

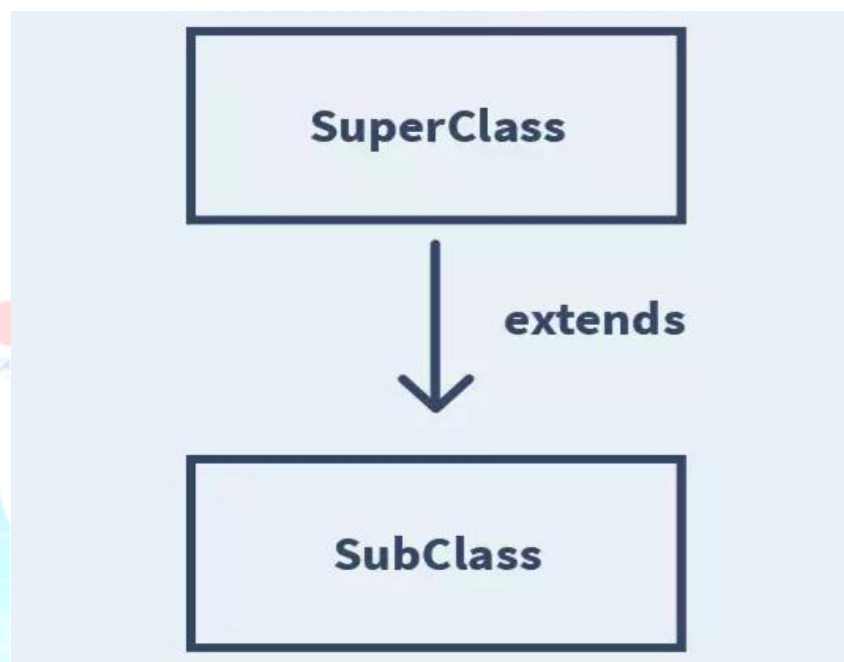
Types of Inheritance in Java:

There are five types of inheritance in java which is given below-



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1. Single Level Inheritance: - This is the simplest form of inheritance, where one class inherits another class.



single level inheritance diagram

2. Multi Level Inheritance: - This is an extension to single level inheritance in java, where another class again inherits the subclass, which inherits the superclass.

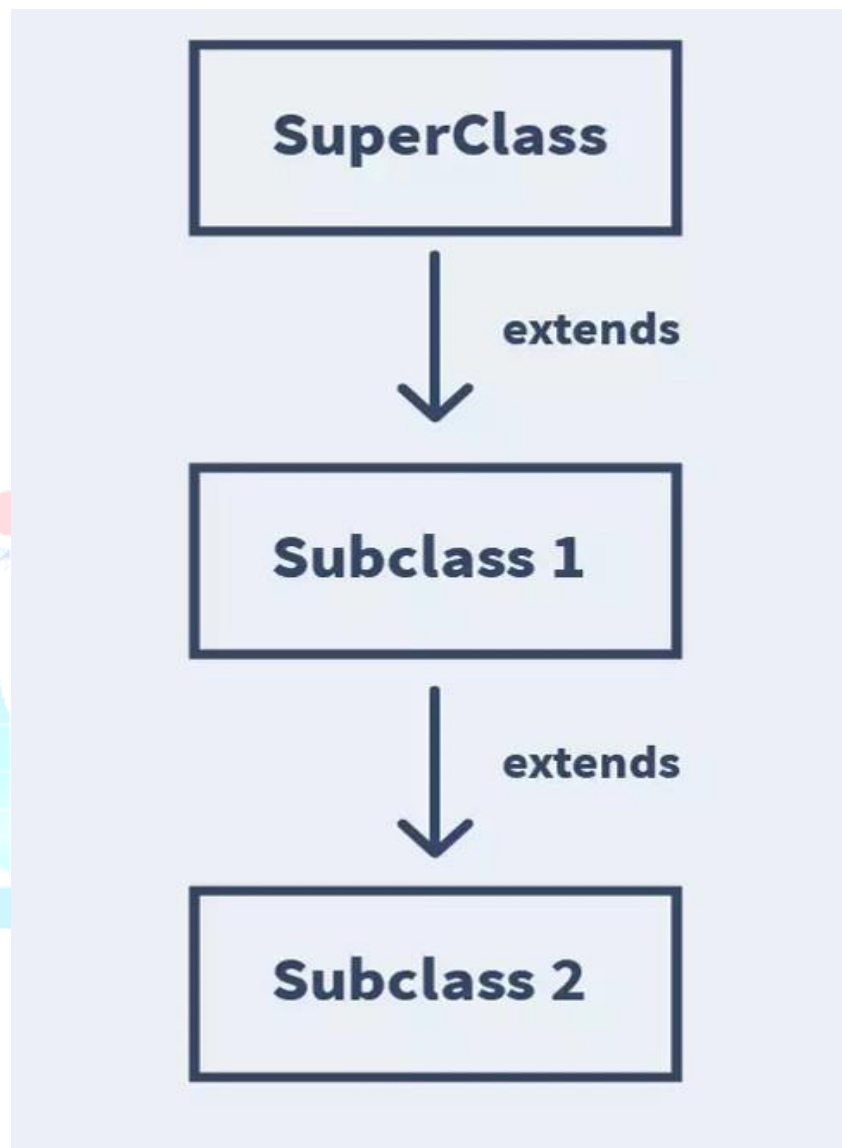
When there is a chain of inheritance, it is also known as Multilevel inheritance.



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multi level inheritance diagram

3. Hierarchical Inheritance: - In hierarchical inheritance, a single superclass is inherited separately by two or more

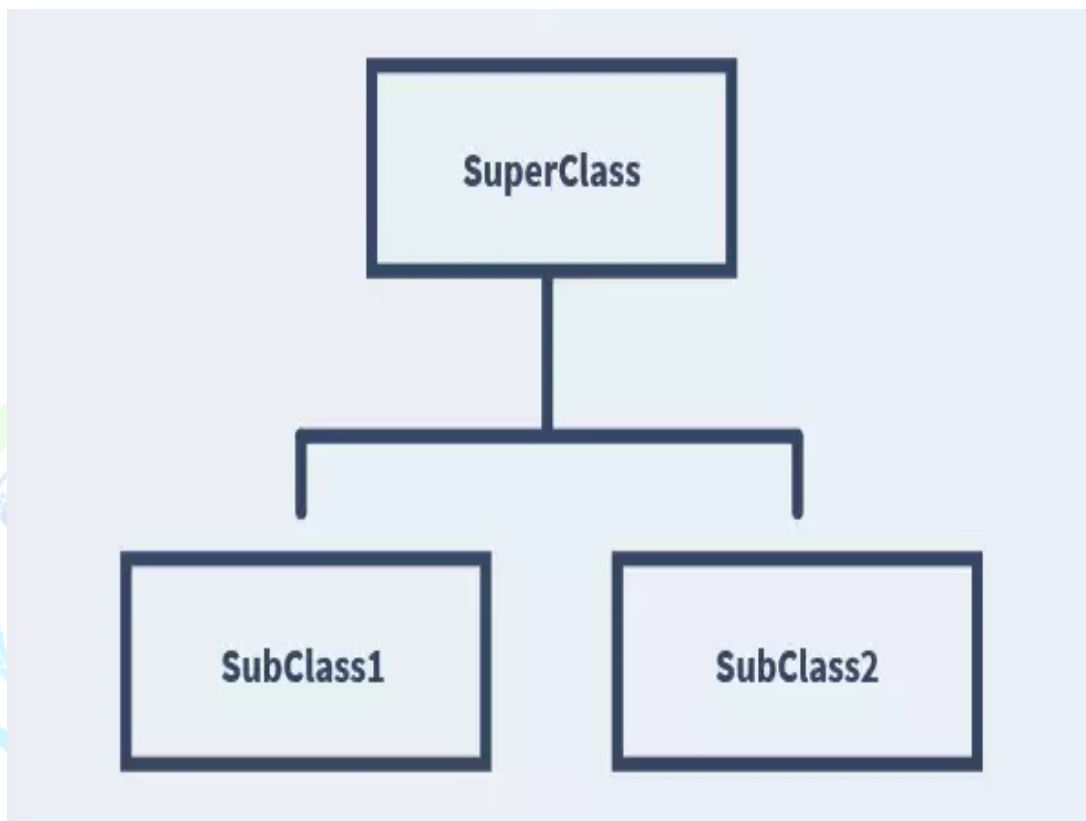


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subclasses.



hierarchical inheritance diagram

4. Multiple Inheritance: - In multiple inheritance, a single class inherits from two different super-classes.

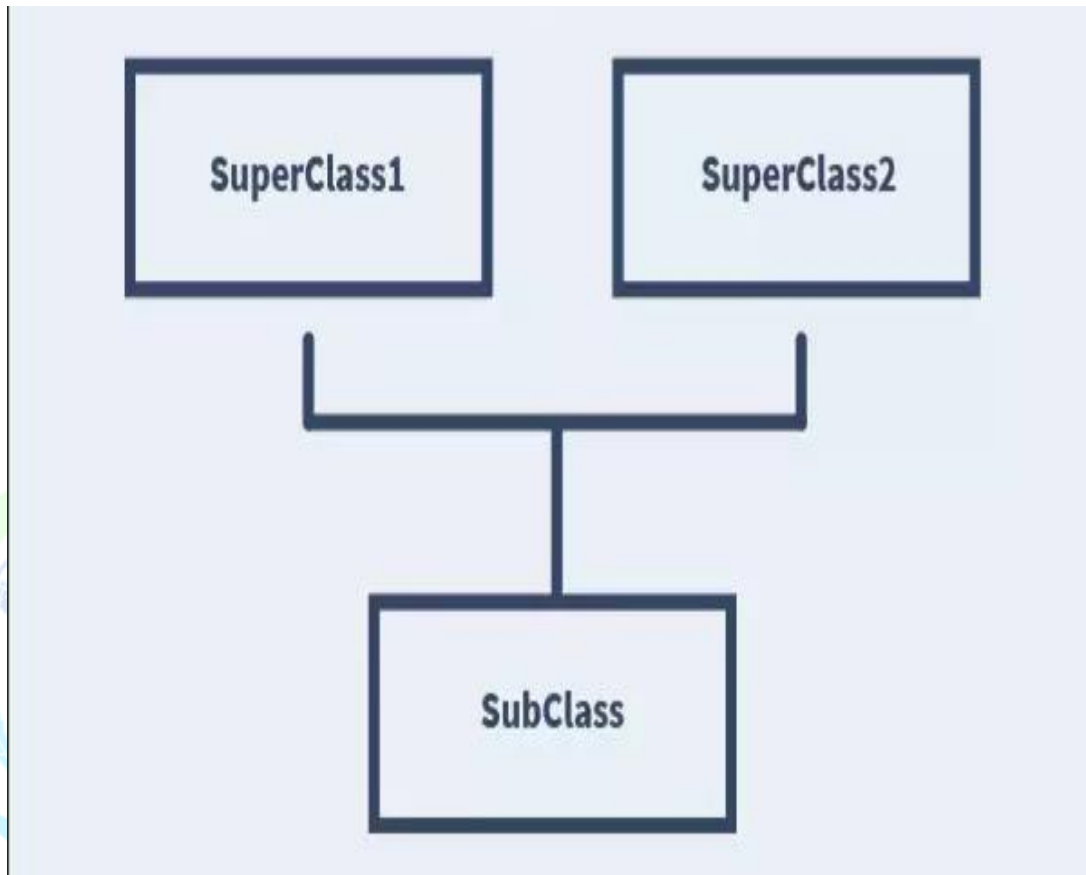
Multiple inheritance is not supported by **java class**.



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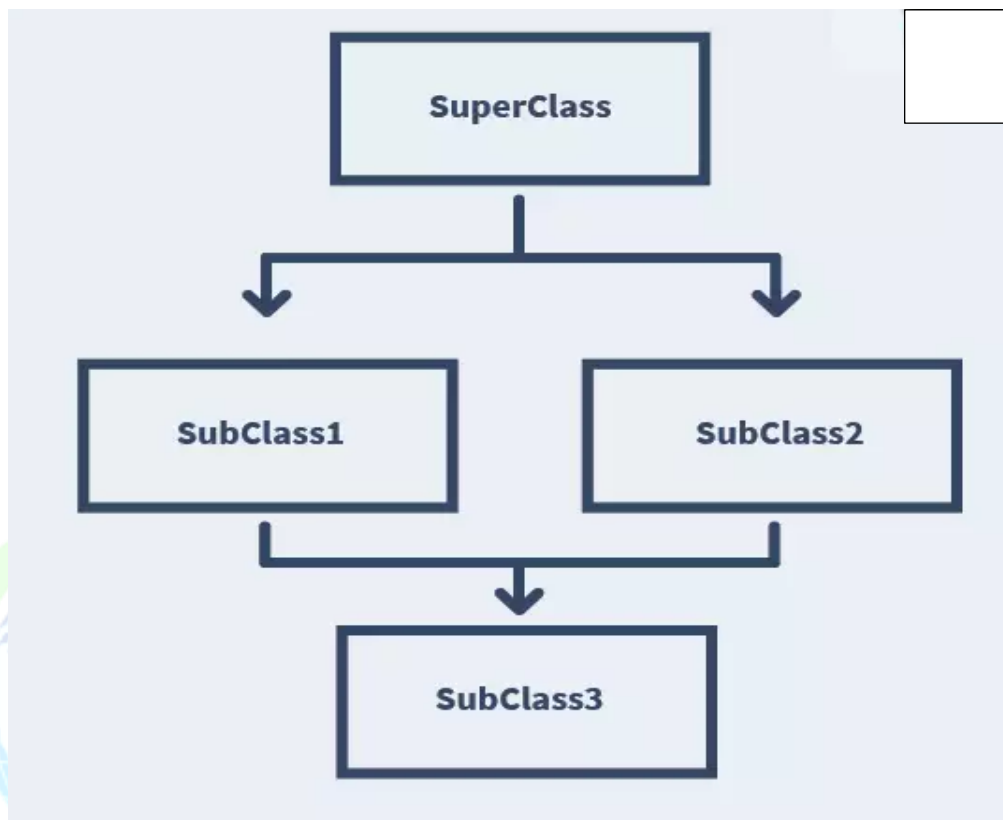
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multiple inheritance diagram

5. Hybrid Inheritance: - In multiple inheritance is a combination of hierarchical inheritance and multiple inheritance.

Multiple inheritance is not supported by java class that's why hybrid inheritance is not possible in java class.



hybrid inheritance diagram

Point to be Remember:

1. Constructor can not be inherited in java.
2. Private member does not get inherited in java.
3. Assign parent reference to child objects.
4. Constructor get executed because of super () present in the constructor.
5. Cyclic inheritance is not permitted in java.